Arnay Garhyan

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Dedicated, responsible, analytical and a self-starter student with a passion for science, technology, and music.

Academics

- GPA (as of end of Junior year): 4.000 unweighted, 4.372 weighted
- SAT: 1600 (800 Reading and Writing, 800 Math), 1 attempt in December 2024
- AP Exams:
  - Freshman Year: AP Computer Science A - 5
  - Sophomore Year: AP Calculus BC - 5, AP Calculus BC: AB Subscore - 5
  - Junior Year: AP United States History - 5, AP Spanish Language and Culture - 5, AP English Language and Composition - 5, AP Chemistry - 5
  - Senior Year: AP Statistics - 5, AP Physics C: Mechanics - 5 and Electricity and Magnetism - 5, AP Microeconomics - 5 and Macroeconomics - 5, AP English Literature and Composition - 5

Experience and Extracurricular Activities

RESEARCH TO IMPROVE DRUG CARDIAC SAFETY STUDIES 2023 – Present

- Applied mathematical, statistical, and programming knowledge to model the pharmacokinetic and pharmacodynamic profile of the drug Moxifloxacin in Non-Human Primates using R
- Presented pharmacokinetic model and simulation tool at the internationally-renowned ASCPT Annual Meeting in May, 2025 — the only high school student out of graduate students and professionals

VEX ROBOTICS 2018 – Present

- Primary programmer for VRC team 6842K — utilizing C++, engineering, and mathematical concepts.
- Developed a template for beginner Park Tudor Vex teams to learn programming for robotics. Mentor new teams on their robot design and programming on a regular basis.
- 2024-2025: Ranked 14th in the world, Amaze Award at the Vex Robotics World Championship, Indiana State Tournament Champion, 2x Tournament Champion, 9x additional judged awards
- 2023-2024: 4x Tournament Champion, 2x Tournament Finalists, Excellence Award, and 2x Build Award.
- 2022-2023 Season: Tournament Champion, Indiana State Tournament Finalists, Indiana State Create Award, 2023 VEX Robotics World Championship Amaze Award.
- 2018-2022 (Team 10775A): 6x Tournament Champion, 6x Skills Champion, 3x Excellence Award, 2x Indiana State Skills Champion, Indiana State Excellence Award, VEX Robotics World Championship Think Award.

APP AND WEBSITE DEVELOPMENT

- Conceptualized, designed, programmed, published, and host an online game named Mr. Redsquare Jumper (www.squarejumper.com). Over 800 unique visitors have played the game.
- Created a website to provide details regarding diseases by using an open source database. Won 2nd place in the CSYA Hackathon (www.squarejumper.com/DiseaseVerse).
- Developed a tool to find statistics about any Vex Robotics team by accessing the RobotEvents API and returning relevant information to the user (www.squarejumper.com/VexScout). Several local Vex teams utilize this tool. Won Silver Medal in the Luddy School of Informatics Computer Science Challenge.
- Developed a tool to analyze track and field statistics, “Ready, Set, GO!”. The website won the Congressional App Challenge
- Created an app to encourage students to learn Python programming — an ongoing project that will be submitted to the Congressional App Challenge as well (www.squarejumper.com/LearnPython1).
- Senior Developer at the 6-employee investment startup LunarBoost

PRESIDENT OF COMPUTER SCIENCE CLUB 2022 – Present

- Qualified for the American Computer Science League (ACSL) All-Stars competition in 2023, 2024, and 2025.
- Co-President of the Park Tudor Computer Science Club, previously head of ACSL education.

MODEL UNITED NATIONS (MUN) 2022 – Present

- Developed public speaking and strategic reasoning skills. Won the Best Delegate Award, the highest award, at the University of Dayton MUN Conference and NAIMUN (the largest MUN conference in the U.S.).
- Won the Verbal Commendation award at the Indiana University MUN Conference. Previous mentor and leader within the club.

MUSIC 2013 – Present

- Play piano, alto saxophone, and tenor saxophone. Currently a member of the Park Tudor Upper School Concert, Pep, and Jazz Bands. Volunteer to perform at retirement communities with the student-led nonprofit Jazz for Discovery.

Education

Incoming Freshman at Stanford University, Palo Alto, California (*Freshman to Present)* - Computer Science

Senior at Park Tudor School, Indianapolis, Indiana (*9th Grade to 12th Grade)* - 4.0 Current GPA

- Summa Cum Laude inductee (4.0 GPA, unweighted)

Wolfram Summer Research Program (WSRP) (Summer, 2025)

- Attended the WSRP in the summer after Junior year in Boston and conducted research on an automated insulin dosing strategy in people with Type 1 diabetes
- See Computational Essay here: https://community.wolfram.com/groups/-/m/t/3499212
- Gained mastery of the Wolfram language, advanced programming and problem-solving skills, and explored the power of modern computation. The project was selected by Stephen Wolfram in-person
- Invited to partake in the highly selective Wolfram Emerging Leaders Program (WELP) and to become a Teaching Assistant at next year's WSRP (approximately 15 of the 75 students)

Sycamore School, Indianapolis, Indiana (*Pre-K to 8th Grade)*

Certifications and Technical Skills

- Python: Certified Entry (PCEP) and Associate Level (PCAP) Programmer
- R, HTML, JavaScript, CSS, C++, Java, Visual Studio Code, GitHub, and Jupyter
- Autodesk Fusion 360